

THE 5x5 PHYSICAL AGENT CO-DESIGN MATRIX

The Master Systems Synthesis: Balancing Physics, Silicon, Memory, Models, and Authority

1. Physical Column	2. Silicon Substrate	3. Algorithmic Organ	4. Failure Mode	5. Systems Invariant
TIME	Cortex MPU ↔ RT MCU	1 kHz QP Barrier Solver	Tail latency overrun	P99.9 Wall Latency < Budget
INERTIA	Static SRAM (Zero Malloc)	Action Chunking (H=16)	Kinetic collision	$d_{stop}(t) \leq d_{clearance}$
ACTUATION	Gate Drivers / Inverters	C ² Quintic Trajectory	Gearbox jerk shock	Jerk Continuity $j \leq j_{max}$
ENERGY	Dynamic Voltage Scaling	Multi-Rate Observers	Thermal throttling	Junction Temp $T_j \leq 85^{\circ}\text{C}$
SILICON	Lock-Free DMA TCM	Expiring Intent Leases	Uncontained crash	Watchdog Lease $\tau \leq 50\text{ ms}$

THE CO-DESIGN AXIOM

Optimizing any single row in isolation without satisfying all 5 columns guarantees physical failure in closed-loop operation.